

Scoring Point-Cloud Distributional Submissions

The deconvolution incentive, jittered outcomes, and the heat ladder

Peter Cotton · Working draft v0.1 · 2026

Status. An evolving working note, not a finished paper. Theorems 1–3 are proved below at working-draft rigor; the mechanism and the closed forms are implemented and unit-tested in [mechanisms/nearest_the_pin.py](#) (mollified_log_score, and tests that verify Theorem 1 numerically in both directions). The prior-art audit lives in [research/mollified-scoring-and-the-heat-ladder.md](#). Drafted with AI assistance; errors are mine.

Abstract

Forecasting contests increasingly accept **point clouds** — bags of Monte-Carlo samples standing in for a predictive density — and score them by smoothing the cloud into a density (a KDE) and evaluating at the realised outcome. We show this common design is **improper**: because the mechanism smooths the submission, the optimal cloud is drawn not from the forecaster’s belief but from its **deconvolution** by the smoothing kernel (Theorem 1; for a Gaussian belief $N(\mu, \tau^2)$ and bandwidth h , the optimal submission is $N(\mu, \tau^2 - h^2)$ — shave exactly the bandwidth off your variance), with a truthfulness gap equal to $\text{KL}(p^* \| p^* * \varphi_h) > 0$. The repair is symmetric and one line: **if you smooth the forecasts, smooth the outcome too**. Jittering the outcome by the same kernel — equivalently, the *mollified log score* $S_h(\rho, z) = (\varphi_h * \log \rho_h)(z)$ — is strictly proper for the pre-smoothing law, because convolution by a kernel whose characteristic function has dense support is injective on probability measures (Lemma 1, Theorem 2). Running the proper rung at every smoothing scale then turns de Bruijn’s identity into a payment schedule (Theorem 3): rung differences pay the Fisher divergence — normalization-free elicitation of *shape* — while the coarse rung pays for between-mode *mass*, and the telescoped total is the full log-score edge. The resulting **heat-ladder pool** is budget-balanced and strictly proper rung by rung. A historical note records that the microprediction platform jittered submissions and ground truth from its launch — a purely intuitive choice at the time, with no theorem behind it; Theorem 2 is that theorem, and it sharpens the intuition into kernel guidance.

1. The problem

A contest accepts from each participant i a cloud $x_1^{(i)}, \dots, x_m^{(i)} \in \mathbb{R}^d$, smooths it into a density $\hat{q}_i = \frac{1}{m} \sum_j \varphi_h(\cdot - x_j^{(i)})$, and — when the outcome z is revealed — rewards $\log \hat{q}_i(z)$ (or splits a pot in proportion to $\hat{q}_i(z)$, the exponentiated form). This is the reward rule of the [nearest-the-pin parimutuel](#) and of monteprediction-style contests, and the standard evaluation protocol for sample-based generative models.

The question this paper answers: *what cloud should a rational participant submit?* The answer is **not** “samples from your belief,” and the failure has a clean closed form, a one-line repair, and — once repaired — a multi-scale structure worth wanting for its own sake.

Throughout, we idealise the cloud by its sampling law ρ ($m \rightarrow \infty$; finite m is Open Problem 1), so the mechanism observes $T_h \rho := \rho * \varphi_h$, where φ_h is a density on $\mathbb{R}^d$, symmetric, strictly positive (the Gaussian $N(0, h^2 I)$ is canonical). The truth is p^* ; all expectations below are assumed finite (for Gaussian kernels and beliefs with finite second moments, $\log T_h \rho$ has at-worst-quadratic tails, so this is mild).

2. The deconvolution incentive

Theorem 1 (improperness of raw-outcome KDE scoring). *Let $S_{\text{raw}}(\rho, z) = \log(T_h \rho)(z)$. Then*

$$\mathbb{E}_{z \sim p^*} S_{\text{raw}}(\rho, z) = -H(p^*) - \text{KL}(p^* \| T_h \rho),$$

so the report solves $\min_{q \in T_h \mathcal{P}} \text{KL}(p^* \| q)$, where $\mathcal{P}$ is the set of probability laws. Consequently:

(i) If the deconvolution $\rho^\dagger := T_h^{-1} p^*$ exists in $\mathcal{P}$, it is the unique optimal report (uniqueness by Lemma 1), and it differs from the truth.

(ii) Truthful reporting $\rho = p^*$ is strictly suboptimal, with truthfulness gap

$$\Delta = \text{KL}(p^* \| p^* * \varphi_h) > 0 \quad \text{for every } h > 0.$$

(iii) Gaussian closed form: $p^* = N(\mu, \Sigma)$, $\varphi_h = N(0, h^2 I)$. The deconvolution exists iff $\Sigma - h^2 I \succeq 0$, and then $\rho^\dagger = N(\mu, \Sigma - h^2 I)$. In one dimension: shave h^2 off the variance, with gap $\Delta = \frac{1}{2}[\log(1 + h^2/\tau^2) - h^2/(\tau^2 + h^2)] \approx h^4/(4\tau^4)$.

Proof. The identity is the standard cross-entropy decomposition, $\int p^* \log q = -H(p^*) - \text{KL}(p^* \| q)$, applied to $q = T_h \rho$; maximizing over ρ is minimizing KL over the convex image class $T_h \mathcal{P}$. (i) If $p^* \in T_h \mathcal{P}$ the KL term can be driven to zero, its unique minimum, and only by $q = p^*$; Lemma 1 lifts uniqueness of q to uniqueness of ρ . (ii) The truthful report attains $\text{KL}(p^* \| p_h^*)$ where $p_h^* := p^* * \varphi_h$; this is strictly positive unless $p_h^* = p^*$. Taking characteristic functions, $\hat{p}^*(\omega)(1 - \hat{\varphi}_h(\omega)) = 0$ for all ω ; for the Gaussian kernel $\hat{\varphi}_h(\omega) = e^{-h^2|\omega|^2/2} < 1$ off $\omega = 0$, so $\hat{p}^*$ would have to vanish off the origin, impossible for a characteristic function (continuity and $\hat{p}^*(0) = 1$). (iii) Gaussians: convolution adds covariances; the KL between $N(0, \tau^2)$ and $N(0, \tau^2 + h^2)$ is the stated expression; Taylor expansion gives $h^4/(4\tau^4)$. ■

Remarks. (a) When the deconvolution fails to exist ($\Sigma - h^2 I \not\succeq 0$, or a rough p^*), the optimum is the KL projection of p^* onto $T_h \mathcal{P}$ and can degenerate toward atomic clouds — exactly the point-mass exploit of Theis, van den Oord & Bethge (2016), who observed the improperness (without the deconvolution characterization). (b) The gap is fourth order in h , which is why the flaw survives casual inspection — but it is a *slope*, and any optimising submitter, human or fitted, walks down it. (c) The phenomenon is the log-score/bandwidth sibling of the “fair scores” findings for the ensemble CRPS (Fricker, Ferro & Stephenson 2013; Ferro 2014), where the analogous cheat is derived along the ensemble-size axis.

3. The repair: jitter the pin

Definition (mollified log score). With $\varepsilon \sim N(0, I)$,

$$S_h(\rho, z) = \mathbb{E}_\varepsilon[\log(T_h\rho)(z + h\varepsilon)] = (\varphi_h * \log T_h\rho)(z).$$

Operationally: *settle at a jittered outcome* $z' = z + h\varepsilon$ — or use the right-hand form, which integrates the jitter analytically and makes settlement deterministic. The two have the same expectation, hence identical incentive properties.

Lemma 1 (injectivity criterion). *Convolution T_φ is injective on probability measures iff $\{\hat{\varphi} \neq 0\}$ is dense in $\mathbb{R}^d$.*

Proof. $(\Leftarrow)$ $T_\varphi\rho = T_\varphi\rho'$ gives $(\hat{\rho} - \hat{\rho}')\hat{\varphi} \equiv 0$, so $\hat{\rho} = \hat{\rho}'$ on a dense set, hence everywhere by continuity of characteristic functions, hence $\rho = \rho'$ by the uniqueness theorem. $(\Rightarrow)$ If $\hat{\varphi}$ vanishes on an open ball B , take a real, even, nonzero g with $\hat{g} \in C_c^\infty(B \cup (-B))$ (so g is Schwartz, $\int g = 0$) and a base density p with tails heavier than Schwartz decay (Cauchy). Then $q = p + \epsilon g$ is a probability density distinct from p for small $\epsilon > 0$, and $\widehat{T_\varphi q} = \hat{\varphi}(\hat{p} + \epsilon\hat{g}) = \hat{\varphi}\hat{p}$ since $\hat{g}$ lives where $\hat{\varphi} = 0$: two distinct laws with identical smoothings. ■

This is Wiener-flavoured (Wiener’s Tauberian theorem is the L^1 statement); for probability measures the continuity of characteristic functions buys the dense-support version. Practical reading: **Gaussian and Laplace jitter are injective** (nonvanishing characteristic functions); the **uniform kernel is also fine** (sinc zeros are isolated, hence the nonvanishing set is dense); **band-limited kernels fail** (e.g. Fejér-type kernels, whose characteristic function vanishes on a half-line — two beliefs agreeing on low frequencies become indistinguishable).

Theorem 2 (kernel-channel properness). *Let S be a scoring rule, K a Markov kernel with push-forward T_K on laws, and define*

$$S_K(\rho, z) = \mathbb{E}_{z' \sim K(z, \cdot)}[S(T_K\rho, z')].$$

If S is proper, S_K is proper. If S is strictly proper on the image class $T_K\mathcal{P}$, then S_K is strictly proper on $\mathcal{P}$ iff T_K is injective on $\mathcal{P}$. In particular the mollified log score S_h is strictly proper for the pre-smoothing law whenever $\{\hat{\varphi} \neq 0\}$ is dense — for Gaussian jitter, always.

Proof. By Fubini, $\mathbb{E}_{z \sim p^*} S_K(\rho, z) = \mathbb{E}_{w \sim T_K p^*} S(T_K\rho, w)$: the expected score of the report $T_K\rho$ for the outcome law $T_K p^*$, under S . Properness of S maximizes this at $T_K\rho = T_K p^*$, which the truthful $\rho = p^*$ achieves — properness. Strict properness of S on the image class forces $T_K\rho = T_K p^*$ at any maximizer; injectivity of T_K lifts this to $\rho = p^*$, and conversely if T_K is not injective two distinct reports tie. For $S = \log$ and $K = \text{convolution by } \varphi_h$: the log score is strictly proper on densities, $T_K\rho = \rho * \varphi_h$ is a density, and Lemma 1 gives injectivity. ■

How much to jitter? Exactly as much as you smooth. The jitter is not a free parameter: it is pinned, kernel for kernel, to the mechanism’s own smoothing. Jittering with s.d. j while smoothing with bandwidth h pairs the jittered truth $p^* * \varphi_j$ with the smoothed report $\rho * \varphi_h$, and in the all-Gaussian setting the optimal report variance is

$$v^* = \tau^2 + j^2 - h^2.$$

At $j = 0$ this is Theorem 1’s shave; at $j = h$ — and only at $j = h$ — the optimum is the truth; at $j > h$ the mechanism pays *padding* by $j^2 - h^2$. Under-jitter rewards sharpening, over-jitter

rewards blurring, and matching the bandwidth is the unique fixed point (the general statement is Theorem 2 with the *same* kernel on both sides; a mismatched pair elicits $\arg \min_{\rho} \text{KL}(p^* * \varphi_j \| \rho * \varphi_h)$, the j -blurred belief deconvolved by h). If the KDE bandwidth is data-driven (Scott’s rule, say), the pin is jittered with that same realised bandwidth.

Attribution, precisely. The *properness* of the convolved score against a noised outcome is not new: Bröcker & Smith (2007, §5) prove it for a general observation-noise channel, and Ferro (2017, Prop. 3) works out exactly the white-noise/Gaussian case. What both left open is *strictness* — Bröcker & Smith, verbatim: “If S is strictly proper though, $\bar{S}$ is not necessarily strictly proper, because if $\bar{q}(z) = \bar{p}(z)$, this does not necessarily mean equality of $p(x)$ and $q(x)$.” Lemma 1 closes exactly that gap. The discrete-outcome analog is the label-noise literature’s forward loss correction with an invertible transition matrix (Patrini et al. 2017, Thm. 2; van Rooyen & Williamson 2018). The framing flip is the contribution of this section: verification treats outcome noise as a *nuisance to be endured* (Ferro explicitly doubts the “efficacy” of perturbing observations); read as *mechanism design*, deliberately injecting jitter matched to the mechanism’s own smoothing is what makes the point-cloud game strictly proper.

4. The heat ladder

Let $p_t := p * N(0, tI)$, the heat flow ($\partial_t p_t = \frac{1}{2} \Delta p_t$). For clouds this is free to compute: the rung at scale t is *the same cloud* scored with bandwidth $\sqrt{h^2 + t}$ against a correspondingly jittered pin, i.e. the §3 score with kernel $\varphi_{\sqrt{h^2 + t}}$.

Theorem 3 (scale decomposition of the log-score edge). *Let p^*, ρ have finite second moments and densities that are positive and smooth after any positive smoothing (automatic here, since every rung applies $t \geq h^2 > 0$ of Gaussian smoothing to both). Then for $0 \leq s < T$,*

$$\frac{d}{dt} \text{KL}(p_t^* \| \rho_t) = -\frac{1}{2} D_F(p_t^* \| \rho_t), \quad \text{KL}(p_s^* \| \rho_s) = \text{KL}(p_T^* \| \rho_T) + \frac{1}{2} \int_s^T D_F(p_t^* \| \rho_t) dt,$$

where $D_F(p \| q) = \int p \|\nabla \log p - \nabla \log q\|^2$ is the Fisher divergence.

Proof. Write $p = p_t^*, q = \rho_t$, both solving $\partial_t u = \frac{1}{2} \Delta u$. Then

$$\frac{d}{dt} \int p \log \frac{p}{q} = \int (\partial_t p) \log \frac{p}{q} + \underbrace{\int \partial_t p}_{=0} - \int \frac{p}{q} \partial_t q.$$

First term: $\frac{1}{2} \int \Delta p \log(p/q) = -\frac{1}{2} \int \nabla p \cdot \nabla \log(p/q) = -\frac{1}{2} \int p \nabla \log p \cdot (\nabla \log p - \nabla \log q)$.

Third term: $-\frac{1}{2} \int \frac{p}{q} \Delta q = \frac{1}{2} \int \nabla \left(\frac{p}{q} \right) \cdot \nabla q = \frac{1}{2} \int p (\nabla \log p - \nabla \log q) \cdot \nabla \log q$. Summing,

$$\frac{d}{dt} \text{KL} = -\frac{1}{2} \int p \|\nabla \log p - \nabla \log q\|^2 = -\frac{1}{2} D_F(p \| q),$$

and integrating over $[s, T]$ gives the display. The integrations by parts are justified because after smoothing by $t \geq h^2 > 0$ both densities are positive, smooth, with Gaussian-dominated tails and integrable score functions; boundary terms vanish. ■

The differential identity is de Bruijn’s, in relative form (Stam 1959; Barron 1986; Lyu 2009); its integral form prices the likelihood of diffusion models (Song, Durkan, Murray & Ermon 2021). *The identity is not ours; the mechanism reading is.* Note the bandwidth floor does real work: at $t = 0$ an empirical cloud has no density and the identity is vacuous; every rung the mechanism actually runs starts at $t \geq h^2$, where everything is smooth.

The heat-ladder pool. Fix scales $0 = t_0 < t_1 < \dots < t_K = T$ and weights $w_k \geq 0$. Participant i stakes s_i and submits one cloud. At settlement, rung k splits $w_k \sum_i s_i$ by any budget-balanced rule driven by the rung score $S_{\sqrt{h^2+t_k}}(\rho_i, z)$ — the stake-weighted additive form $\Delta W_i^{(k)} = w_k s_i (S_i^{(k)} - \bar{S}^{(k)})$ with $\bar{S}^{(k)}$ the stake-weighted mean (Lambert et al. 2008), or the multiplicative pot split at full Kelly.

Corollary. (i) *Each rung, hence the tower, is budget-balanced.* (ii) *Each rung is strictly proper for the cloud law (Theorem 2), so truthful submission is optimal rung-wise in the small-stake, risk-neutral limit.* (iii) *By Theorem 3 a truthful participant’s edge decomposes across rungs: differences of adjacent rungs pay (integrated) Fisher divergence — Hyvärinen-scored shape, invariant to the normalization of the submission — while the top rung pays $\text{KL}(p_T^* \| \rho_T)$, which at mode-connecting scales carries the between-mode mass that score matching is blind to (Wenliang & Kanagawa 2020; Zhang et al. 2022; Koehler, Heckett & Risteski 2023). The weights w_k are an explicit dial over what is being purchased.*

Practicalities: rung scores are positively correlated (one cloud, re-smoothed), so discriminative value concentrates in a few well-separated scales; K of order 3–5, geometrically spaced, mirrors the noise ladders of annealed score matching (Song & Ermon 2019).

5. Historical note: the jitter was intuition first

The microprediction platform (Cotton 2022; launched 2019) added a small amount of noise to submissions and to the ground truth before settling its cloud-based lotteries. **This was merely intuitive** — a fairness-and-anti-gaming instinct about discreteness and ties, with no incentive theorem attached; the platform paper recorded the practice in one line and moved on. The verification literature, meanwhile, treated outcome noise as a defect: something to be modelled away (Saetra et al. 2004; Candille & Talagrand 2008), with Ferro (2017) explicitly doubting the value of perturbing observations.

Theorem 2 is the missing theorem: symmetrized jitter is precisely what makes a smoothed-submission game strictly proper for the submitted law. And the theory returns the favour by *sharpening* the intuition into design guidance the intuition could not supply: the jitter distribution matters. Kernels whose characteristic function has dense support (Gaussian, Laplace, even uniform) preserve strict properness; band-limited kernels do not (Lemma 1). Intuition chose jitter; the theorem chooses *which* jitter.

6. Related work

Improperness of sample-based scoring: Theis, van den Oord & Bethge (2016) (KDE log-likelihood “an improper scoring function”); the fair-scores line for the ensemble CRPS (Bröcker

2012; Fricker, Ferro & Stephenson 2013; Ferro 2014); the estimator view of KDE log scores (Krüger, Lerch, Thorarinsdottir & Gneiting 2021); discrete impossibility and randomized repair (Kimpapa, Frongillo & Waggoner 2023). Convolved scores under observation error: Bröcker & Smith (2007); Ferro (2017); Bessac & Naveau (2021). Noisy-channel learning: Patrini et al. (2017); van Rooyen & Williamson (2018); surrogate scoring rules (Liu, Wang & Chen 2022). Transform-properness: Allen, Ginsbourger & Ziegel (2023, Prop. 4 published / Prop. 3 arXiv v1); Pic, Dombry, Naveau & Taillardat (2025, Prop. 1) — deterministic injective transforms; Theorem 2 is the Markov-kernel extension. The identity: Stam (1959); Barron (1986); Lyu (2009); Song et al. (2021). Local scores: Hyvärinen (2005); Parry, Dawid & Lauritzen (2012). Nearest mechanism neighbours, each missing a leg: Lang, Leutbecher & Maciel (2025) — a Gaussian scale-ladder of proper scores as a *training loss*; Dudík, Wang, Pennock & Rothschild (2021) — a multi-resolution *market* by partition refinement, subsidized rather than budget-balanced; Lambert et al. (2008, 2015) and the microprediction pool — self-funding cloud wagering at a single scale. A full audit with verdicts is in the companion [research note](#).

7. Open problems

1. **Fair rungs.** The finite- m correction making each rung’s expected score optimized by *sampling* from the belief (the log-score/KDE analog of Ferro’s fair CRPS), and its interaction with the jitter.
2. **Optimal scale weights.** For wealth-concentration objectives, is there a closed-form optimal $w(t)$ — and does it recover the likelihood weighting of Song et al. (2021)?
3. **Anisotropic rungs.** Reference-covariance flows in place of isotropic heat, preserving rung-wise strict properness (cf. the [anisotropic sliced scores note](#)).
4. **The Kelly interpolation.** Between $b \rightarrow 0$ (linear pot split, mode-seeking) and $b = 1$ (log score, truthful), characterize the rung-wise optimal misreport as a function of the stake fraction.

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